Second adjointness

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Setup. - G split reductive group $/F$ (e.g. $\mathbb{Q}_p$)

- $k = \mathbb{C}$ coeff. ring

Recall. $P = MU$ of G.

we have $i_P := i_M^G : \mathrm{Rep}(M) \longrightarrow \mathrm{Rep}(G)$

$r_P := r_M^G : \mathrm{Rep}(G) \longrightarrow \mathrm{Rep}(M)$

adjoint pair (r_P, i_P)

Now we consider $\bar{P} = M \cdot \bar{U}$ opposite parabolic.

$r_{\bar{P}} = \overline{r_M^G}$

Thm A. i_P is left adjoint to $r_{\bar{P}}$, i.e. $\mathrm{Hom}_G(i_P(V), W) \Rightarrow \mathrm{Hom}_M(V, r_{\bar{P}}(W))$.

Thm B. $V \in \mathrm{Rep}(G)$, $r_{\bar{P}}(\tilde{V}) \overset{\text{canonically}}{\simeq} \widetilde{r_P(V)}$.

(Recall: $\widetilde{i_P(V)} = i_P(\tilde{V})$) $\qquad \widetilde{(-)}$: smooth dual

Claim. Thm A $\Leftrightarrow$ Thm B.

$\forall\, T \in \mathrm{Rep}(M)$,

$$\begin{aligned}\mathrm{Hom}_M(T, \widetilde{r_P(V)}) &\simeq \mathrm{Hom}_M(r_P(V), \tilde{T}) \\ &\simeq \mathrm{Hom}_G(V, i_P(\tilde{T})) \\ &\simeq \mathrm{Hom}_G(V, \widetilde{i_P(T)}) \simeq \mathrm{Hom}_G(i_P(T), \tilde{V}) \qquad (*)\end{aligned}$$

Thm A $\Rightarrow$ $\quad \mathrm{Hom}(i_P(T), \tilde{V}) \simeq \mathrm{Hom}(T, r_{\bar{P}}(\tilde{V}))$

$\Rightarrow$ Thm B (Yoneda's lemma)

Thm B $\Rightarrow$ Thm A : $\quad \mathrm{Hom}(i_P(T), \tilde{V}) \simeq \mathrm{Hom}(T, r_{\bar{P}}(\tilde{V}))$ for any T, V

for general $\pi \in \mathrm{Rep}(G)$, $0 \to \pi \hookrightarrow \tilde{\tilde{\pi}} \twoheadrightarrow \widetilde{(\tilde{\tilde{\pi}}/\pi)}$ exact $\square$

Strategy: $\forall\, V \in \mathrm{Rep}(G)$, $\forall$ congruence subgp K

identify

$$r_P(V)^{K_M} \overset{\oplus}{\hookrightarrow} V^K$$

$$r_{\bar{P}}(\tilde{V})^{K_M} \overset{\oplus}{\hookrightarrow} \tilde{V}^K = (V^K)^*$$

nat'l pairing

$$(r_P(V)^{K_M})^* \simeq r_{\bar{P}}(\tilde{V})^{K_M}$$

$$\Rightarrow \quad \widetilde{r_P(V)} \simeq r_{\bar{P}}(\tilde{V})$$

find an operator A (depends on $K, P, \bar{P}$)

$$\underset{\substack{\circlearrowleft \\ A \text{ nilp}}}{V^K \simeq V_0} \oplus \underset{\substack{\circlearrowleft \\ A \text{ invertible}}}{V_*}$$

identify $V_* \simeq r_P(M)^{K_M}$.

$$\left(\begin{array}{l} \tilde{V}^K \simeq \tilde{V}_0 \oplus \tilde{V}_* \\ \tilde{V}_* = r_{\bar{P}}(\tilde{M})^{K_M} \end{array} \right)$$

geometrically: $A \curvearrowright V^K$,

$$\{0\} \longrightarrow \mathbb{A}^1 = \mathrm{Spec}\, k[A] \longleftarrow \mathbb{G}_m$$

with V^K over $\mathbb{A}^1$ and $r_P(V)^{K_M}$ over $\mathbb{G}_m$.

Def. L a k-vect. space. $a \in \mathrm{End}_k(L)$. $\Rightarrow L' := L \Big/ \bigcup_{n=1}^{\infty} \ker(a^n)$

define localisation $L_a := L' \otimes_{k[a]} k[a, a^{-1}]$.

$\lambda \in M$ strictly dominant w.r.t. P, $K \leftarrow$ congruence subgroup

- K_U, K_M invariant under Ad_λ
- $K_{\bar{U}}, K_M$ invariant under $\mathrm{Ad}_{\lambda^{-1}}$
- $P = P_\lambda = \{ g \in G : \mathrm{Ad}_\lambda^n \cdot g$ has a limit $\}$.

① K is normalized by a max. cpt open K_0

② $\forall P' = M'U'$ standard parabolic, $K = K_{U'} K_{M'} K_{\overline{U'}}$, where $K_{U'} = K \cap U'$, ...

For any $V \in \mathrm{Rep}(G)$, $\pi : V^K \longrightarrow r_P(V)^{K_M}$ equivariant (using in good position)

$a(\lambda) = e_K * \varepsilon_\lambda \circlearrowleft V^K$, $\quad A = \lambda \circlearrowleft r_P(V)^{K_M}$.

Lemma. $(r_P(V)^{K_M}, A)$ is the localisation of $(V^K, a(\lambda))$.

Proof. need to show: $\forall x \in r_P(V)^{K_M}$, $\exists n$ large enough s.t.

$$\lambda^n \cdot x \in \mathrm{Im}(\pi)$$

Take any lift $x' \in V^{K_M}$. Choose congruence K' s.t. x' is invariant under K'.

$y = \lambda^n x'$ is invariant under $\lambda^n K'_{\bar{U}} \lambda^{-n}$.

$\Rightarrow$ choose n large enough, y is invariant under $K_{\bar{U}}$.

Set $\tilde{x} = e_{K_U} \cdot y$, $\quad \pi(\tilde{x}) = \pi(y) = \lambda^n x \Rightarrow \lambda^n x \in \mathrm{Im}(\pi)$. $\square$
$\quad = e_{K_U} \cdot e_{K_M} \cdot e_{K_{\bar{U}}} \cdot y$

Jacquet's lemma V admissible $\Rightarrow$ $V^K \xrightarrow{\pi} r_P(V)^{K_M}$ is surjective.

Proof $r_P(V)^{K_M}$ is a clocalisation of V^K, V^K, $r_P(V)^{K_M}$ f.d.

$$\Rightarrow r_P(V)^{K_M} \simeq V^K \Big/ \bigcup_{i=1}^{\infty} \ker(a^n).$$

Goal. drop the admissibility assumption.

Idea: V is always admissible over the Bernstein centre $Z(G)$

Def L v.s. $/k$, $a \in \mathrm{End}_k(L)$. We say that (L, a) is stable, if

$L = \ker a \oplus \underset{\substack{\circlearrowleft \\ a \text{ is invertible}}}{\mathrm{Im}\, a}$ $\quad (\Leftrightarrow \ker a^2 = \ker a,\ \mathrm{Im}\, a^2 = \mathrm{Im}\, a)$

Lemma. (L, a) is stable, then $L_a = L' = L \Big/ \bigcup_n \ker(a^n)$

Proof omitted.

Stabilization Thm $K \subset G$ congruence subgp

Choose λ is strictly dominant w.r.t. P, K.

$\exists$ constant $c(G, K)$ s.t. $\forall$ $V \in \mathrm{Rep}(G)$, $\forall n \geq c(G, K)$.

$\left(V^K, a(\lambda^n)\right)$ is stable.

Cor (strong version of JL)

$\pi: V^K \twoheadrightarrow r_P(V)^{K_M}$

Moreover, $r_P(V)^{K_M} \overset{\oplus}{\hookrightarrow} V^K$, functorially in V.

Thm B.

$$r_p(V)^{KM} \overset{\oplus}{\hookrightarrow} V^K$$

$$r_{\bar{P}}(\tilde{V})^{KM} \overset{\oplus}{\hookrightarrow} (V^K)^* \simeq \tilde{V}^K$$

$\circlearrowleft$

λ^{-1} invertible

λ^{-1} is s-dominant w.r.t. $\bar{P}, K$.

$$\langle -,- \rangle : \quad \underset{\substack{\circlearrowleft \\ a(\lambda)}}{V^K} \times \underset{\substack{\circlearrowleft \\ a(\lambda^{-1})}}{(V^K)^*} \longrightarrow k$$

$$r_p(V)^{KM} \times r_{\bar{P}}(\tilde{V})^{KM} \longrightarrow k$$

$$\Rightarrow \quad r_p(V) \times r_{\bar{P}}(\tilde{V}) \longrightarrow k$$

Proof of stabilization thm.

Easy facts: Stable modules form an abelian category

$$f: V \longrightarrow W, \; V, W \text{ stable} \Rightarrow V \oplus W, \ker(f), \operatorname{coker}(f) \text{ stable}.$$

Lemma: B comm. noetherian ring, L f.g. B-module, $a \in \operatorname{End}_B(L)$,

L_a f.g. B-module $\Rightarrow$ $\exists n$ s.t. (L, a^n) is stable.

Proof: (L_a, A) localisation of (L, a), $A^{-1} \in \operatorname{End}_B(L_a)$. $\Rightarrow A^{-1} \in B[A]$.

$L' \subset L_a$ is A-invariant $\Rightarrow$ L' is A^{-1}-invariant $\Rightarrow$ $L' = L_a$. $\square$

L f.g. choose n s.t. $L' = L / \ker(a^n)$, $L = \ker(a^n) \oplus La$. $\square$

$Z(G)$ not noetherian. $Z(G) \simeq \prod Z_s$

ρ fix irred. rep'n. $\mathrm{Rep}(M')_{cusp}$

$D = \{ (\rho, \psi) : \psi \in \Psi(M') \}$

unramified character.

$R = k[\Psi(M')] = \text{c-ind}_{T_M(\mathcal{O})}^{T_M(F)}(\text{triv})$

Define $\pi(D) = R \otimes \rho \in \mathrm{Rep}(M')$

<u>Fact</u> (1) $\pi(D)$ is a projective <u>generator</u> in $\mathrm{Rep}(M')_D$,

$\mathrm{Hom}(\pi(D), -)$ is faithful.

(2) $\pi(D)$ is (M', R)-admissible.

$\Rightarrow i_P(\pi(D))$ is (G, R)-admissible.

$\xRightarrow{\text{Geom. Lemma}} r_P\, i_P(\pi(D))$ is (G, R)-admissible

$\Rightarrow \pi := i_P(\pi(D))$ is stable under a^n (for some n)

$\forall L \in \mathrm{Rep}(M)$, $\prod_\alpha \pi(D)^\alpha \to \prod_\beta \pi(D)^\beta \to L \to 0$

$\prod_\alpha \pi^\alpha \to \prod_\beta \pi^\beta \to i_P(L) \to 0$

$\Rightarrow i_P(L)$ is stable.

$V \in \mathrm{Rep}(G)$, $V = \bigoplus_\Omega V(\Omega)$, $V \hookrightarrow \bigoplus_{M' \sim M} i_{M'}^G(L_{M'})$